LESSON PLAN

COMPOSITE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	11	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Vernier instruments, reading of Vernier scale, types of Vernier Caliper, precautions while using the instrument.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Vernier instruments, reading of Vernier scale, types of Vernier Caliper, precautions while using the instrument.
5. Specific Objectives	19.1.1.1 Trainee will be able to describe details about Engineering Metrology.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Vernier Caliper parts, reading practice on standard gauge blocks, safety check of instrument jaws.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engineering Metrology?
(b) Name one instrument used for measuring internal dimensions.
(c) Why is precision important in aircraft maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principle of Vernier Scale • Least Count Calculation • Reading Vernier Scale • Types of Vernier Caliper • Parts of Vernier Caliper	L & Ds	PPT, IB	15 min
(b)	• Precautions for Use • Handling and Storage • Error Sources • Zero Error Correction • Maintenance Procedures	L & Ds	PPT, PDF	20 min
(c)	• Common Mistakes • Calibration Basics • Record Keeping • Troubleshooting • Safety Protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how to calculate the least count of a Vernier Caliper.
(b) List three precautions to take before using a Vernier Caliper.
(c) Describe the difference between external and internal measuring jaws.

12. Home Assignment and Follow Up	Prepare a brief note on the importance of zero error correction in measurement accuracy for the next lesson on Micrometers.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Micrometer, descriptions of various parts, important terms such as backlash, measuring range, total error, cumulative error: various types of Micrometers.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min